



AGH

AGH UNIVERSITY OF KRAKOW
Faculty of Space Technologies

Remote sensing
Laboratory 7 – Landslides in China

Author:
Graduation path:

Miłosz Niedziątek
downstream

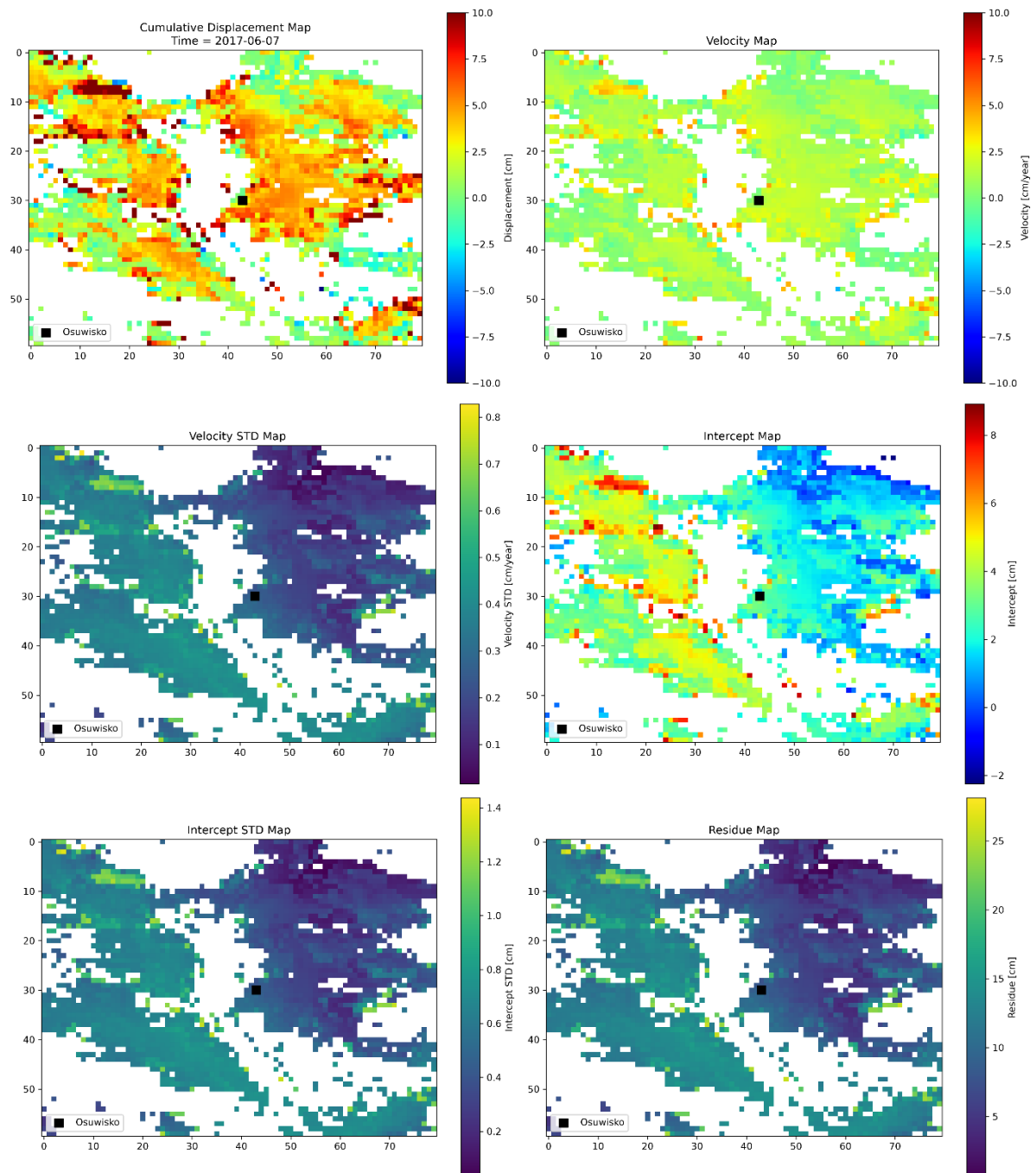
Kraków, 01.07.2026

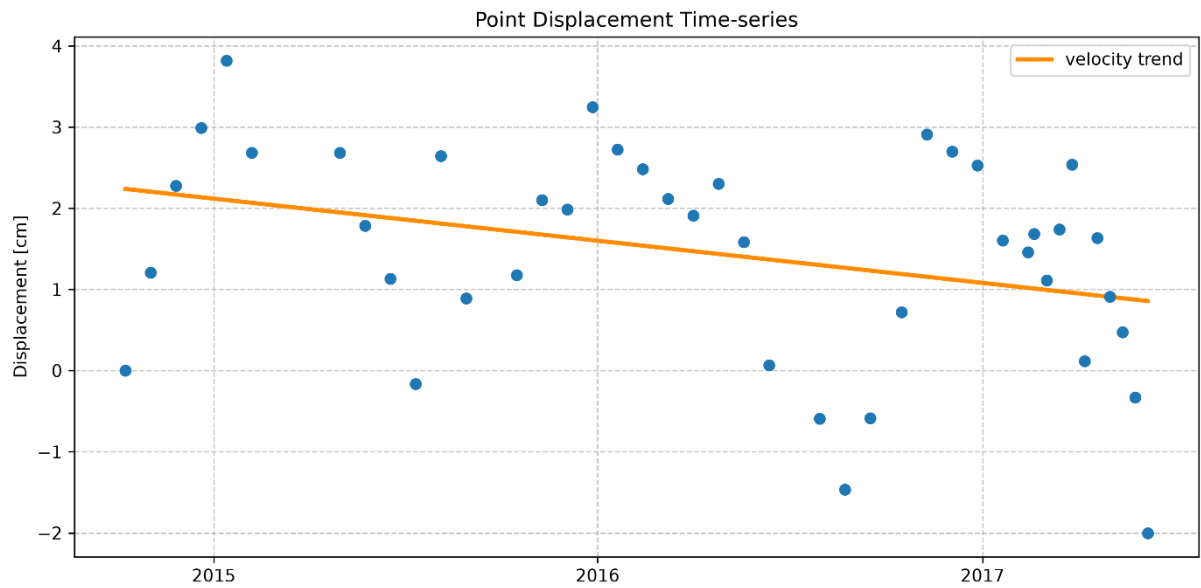
Steps of the Analysis

I downloaded ready-made Sentinel-1 interferograms from the COMET-LiCS portal. To prevent the software from generating excessive noise and crashing during processing, huge satellite frames were cropped down to a small bounding box (103.62-103.70E, 32.05-32.11N) directly over the Xinmo landslide.

SBAS InSAR Analysis (MintPy)

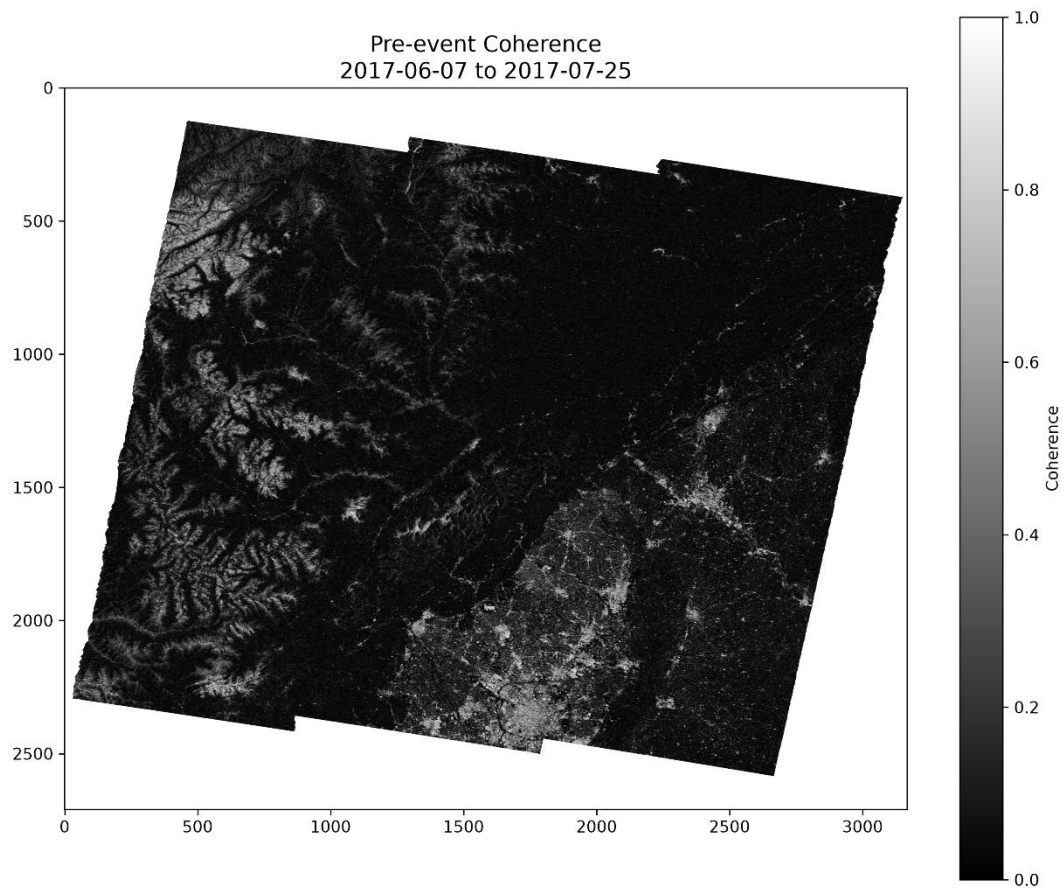
I used the MintPy library to run a Small Baseline Subset SBAS analysis. Because small, forested study area had very low radar coherence, I manually lowered the minCoherence threshold to 0.3 so the algorithm could successfully find a reference point and calculate the deformation time-series.





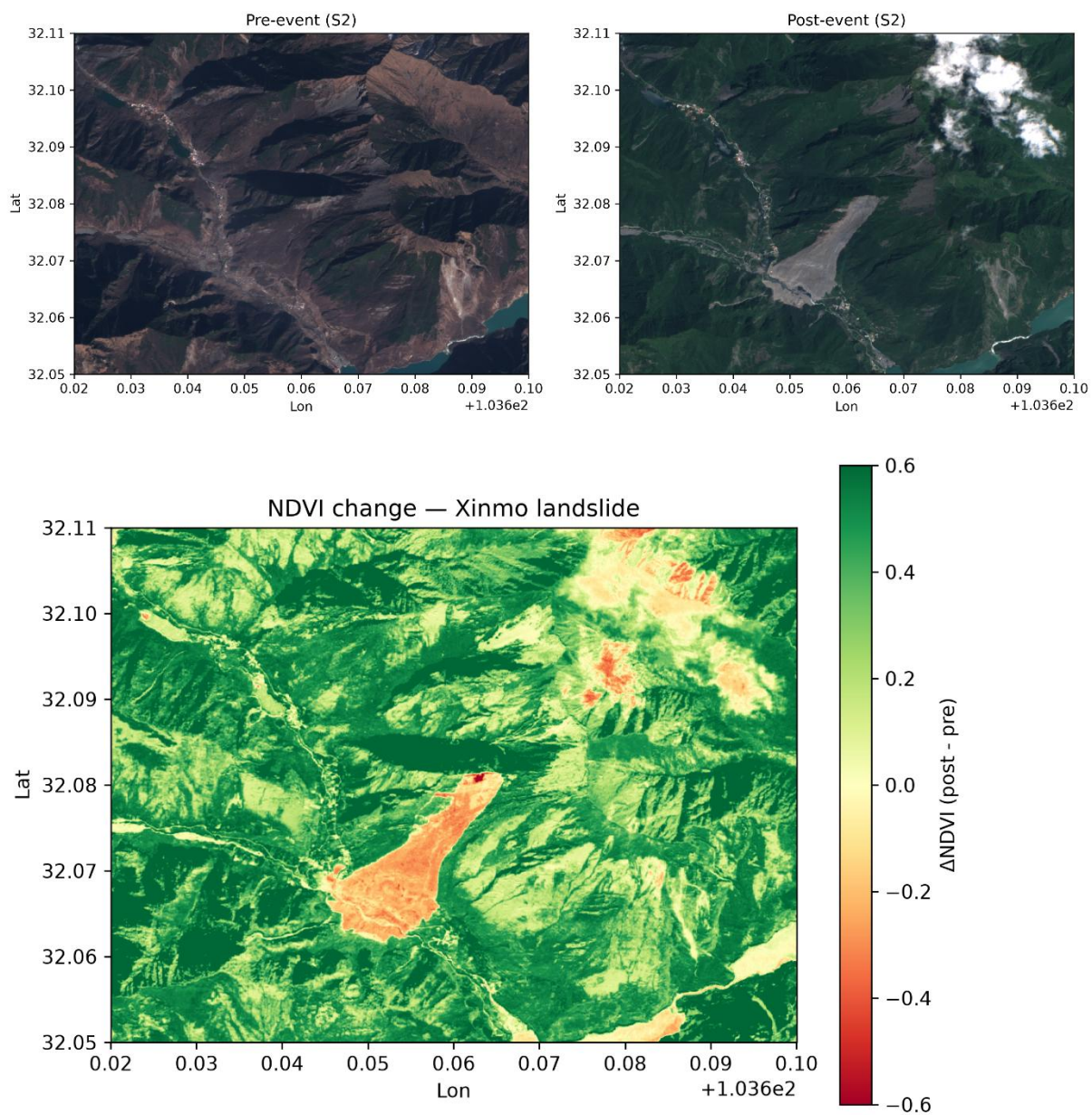
Pre-event Coherence Check

Next I plotted a single pre-event coherence map June–July 2017 to observe the stability of the ground. Dense vegetation and unstable areas appeared as dark, low-coherence pixels, which helped understand the surface conditions before the landslide.



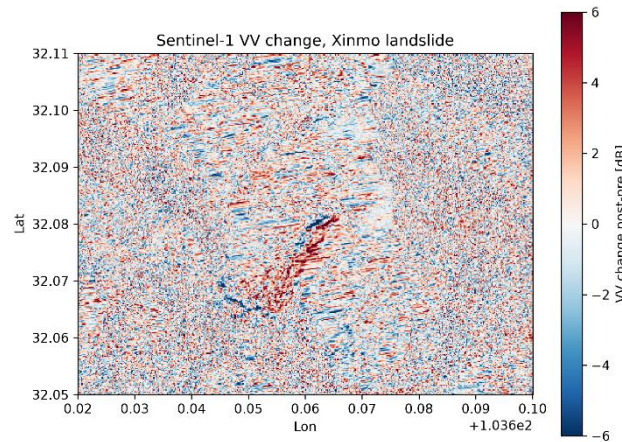
Post-event Optical Analysis (Google Earth Engine)

I calculated the NDVI difference using Sentinel-2 images from before and after the collapse. This highlighted the exact areas where vegetation was destroyed.



Post-event SAR Amplitude Analysis

Finally, we analyzed the change in Sentinel-1 SAR backscatter (VV polarization) using Ascending orbit data, which helped map the rocky debris flow boundaries even further.



Discussion of the Results

The results demonstrated the value of combining different remote sensing techniques to study landslides.

One of the findings came from MintPy InSAR analysis, where the point displacement time-series for the landslide source area revealed a slow, continuous downward trend. Over the years leading up to the catastrophic collapse, the slope had been creeping by a few centimeters away from the satellite sensor.

The mean velocity map also highlighted this specific slope as an active deformation zone compared to the stable mountains surrounding it, proving that InSAR can act as a powerful early warning tool to identify unstable slopes before they fail. However, while InSAR was excellent for detecting this millimeter-scale precursory movement, it became useless for mapping the actual collapse because the radar signal completely decorrelated once the ground gave way.

To solve this, our Google Earth Engine analysis provided the perfect follow-up.

The NDVI change map vividly captured the exact shape, track, and runout distance of the landslide, as the stark contrast between the surviving forest and the bare rock perfectly outlined the path of destruction.

InSAR is ideal for monitoring slow, "sneaky" movements prior to a disaster, while optical indices like NDVI and SAR backscatter changes are the best tools for mapping the true scale of the damage immediately after the event occurs.